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B.Tech. Degree I Semester Examination December 2015

CS/IT/EC 1102 B ENGINEERING CHEMISTRY (2015 Scheme)

Time : 3 Hours

Maximum Marks : 60

PART A (Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Explain different types of electronic transitions.
- (b) Differentiate Schottky & Frenkel defect.
- (c) Give any four applications of salt bridge.
- (d) What is the difference between molecularity and order of a reaction?
- (e) What is Trauton's rule?
- (f) What is eutectic system? Explain.
- (g) Give the method of preparation & important properties of PVC.
- (h) Write any four requisites of refractories.
- (i) What is the difference between injection and extrusion moulding?
- (j) Give the four statements of 2nd Law of thermodynamics.

PART B

(4 × 10 = 40)

- II. (a) Explain fermi level in band theory. (2)
 - (b) Explain thermodynamics of adsorption and BET theorem. (8)
- OR**
- III. (a) Explain the working principle of NMR spectroscopy. (5)
 - (b) What is Beer-Lamberts Law? Explain. (5)
- IV. (a) What is solar cell? Explain the working of solar cell. (5)
 - (b) Derive the Nernst equation. (5)
- OR**
- V. (a) Derive the integral rate law equation for first order reaction. (6)
 - (b) What is the working principle of concentration cell? (4)
- VI. (a) Derive the Gibb's Helmholtz equation. (7)
 - (b) Explain Nernst heat theorem. (3)
- OR**
- VII. (a) Explain Phase Rule taking one component water system as example. (8)
 - (b) What is chemical potential? Explain. (2)
- VIII. (a) Explain the vulcanisation of rubber. (5)
 - (b) Differentiate thermoplastics and thermosets. (5)
- OR**
- IX. Explain the following. (10)
 - (i) Viscosity index.
 - (ii) Flash and fire point.
 - (iii) Cloud and pour point.
 - (iv) Aniline point.

